Supplementary Information

Delta Dependency Profiles Paralog Compensation
Signals Across Solid Tumor Types

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Contents

1 Supplementary Methods

1.1 Curated-pair curation and tier assignment

Each of the twelve curated pairs was verified against its primary citation(s) before inclusion. For every pair we recorded the assay type, model system, validated direction, whether the evidence derives from simultaneous dual-gene perturbation, whether it is independent of DepMap CRISPR data, and whether it directly demonstrates synthetic lethality in the direction scored here (Table S3). Pairs were then assigned to three evidence tiers. Tier A requires direct genetic synthetic-lethal evidence from dual-gene perturbation (combinatorial or digenic CRISPR knockout; 3 pairs). Tier B requires that the paralog be a demonstrated selective dependency in driver-mutant cells from single-gene perturbation with functional validation (natural-genotype conditional dependency; 2 pairs). The Tier A $\cup$ Tier B set constitutes the primary literature-derived benchmark, with an evidence-independence asymmetry: Tier A derives from dual-gene perturbation and is independent of the single-gene dependency data DD measures, whereas Tier B is itself natural-genotype conditional dependency, the same data type as DD, so Tier B concordance is partially expected by construction. Tier C comprises indirect evidence only: reciprocal-direction-only evidence (EP300$\rightarrow$CREBBP), evidence supporting a different driver (PIK3CA$\rightarrow$PIK3CB, whose primary evidence supports PTEN$\rightarrow$PIK3CB), developmental redundancy (CCNE1$\rightarrow$CCNE2), or DepMap computational analyses (FBXW7$\rightarrow$FBXW2, PPP2R1A$\rightarrow$PPP2R1B); it is excluded from the primary benchmark (5 pairs). Two further pairs that are not sequence paralogs (BRCA1$\leftrightarrow$BRCA2, functional analog; STK11$\rightarrow$SIK1, partial homolog/pathway axis) serve as mechanistic comparators and specificity references only. Pharmacologic inhibition of both members of a gene family was not accepted as sufficient evidence of genotype-conditional synthetic lethality in a specified direction.

1.2 Driver-mutation classification

Driver-mutation status was defined per gene class on the default omics profile of each DepMap 26Q1 model (`IsDefaultEntryForModel`). Tumor suppressors (TSG) require a DepMap-curated likely loss-of-function variant (`LikelyLoF`; includes dominant-negative TP53 missense); oncogenes require a recurrent oncogenic hotspot variant (`Hotspot`). Gene classes were assigned by the DepMap 26Q1 `OncogeneHighImpact/TumorSuppressorHighImpact` majority vote, with one documented manual override (GATA3 as TSG). Amplification-driven oncogenes without qualifying hotspot mutations (e.g., CCNE1) contribute no mutant lines under this rule and are analyzed through copy-number channels instead. The per-gene comparison of the permissive non-silent definition versus the class-specific rule, with retained cell line counts and the top VariantInfo terms, is given in Table S6 (machine-readable mirror: `TableS6_MutationRules.tsv`).

1.3 DD computation and evaluation frames

DD was computed per driver$\times$paralog$\times$lineage stratum as the difference in mean Chronos gene-effect scores between driver-mutant and wild-type cell lines (Equation 1 of the main text), with Welch's t-test (unequal variances); effect sizes are reported as Cohen's d (pooled SD) and Hedges' g (small-sample corrected; Table S7). The primary analysis frame requires $\geq$5 mutant and $\geq$5 wild-type cell lines per stratum (110 evaluable driver$\times$paralog$\times$lineage associations in the three evaluation lineages (Ovarian, Endometrial, Cervical) across the 12 curated pairs and matched controls; 17 columns in

TableS2_FullResults.tsv); a relaxed $\geq 3/\geq 3$ frame was re-run as a sensitivity analysis (12 evaluable lineages versus 8 on the primary frame). Benjamini–Hochberg correction ($q < 0.20$) was applied within each driver gene across all of that driver's HGNC paralogs and lineages jointly; a stricter $q < 0.10$ cut was examined as sensitivity. This within-driver correction is a discovery-stage device and does not constitute genome-wide FDR control.

1.4 Classifier training and cross-validation

For the head-to-head comparison on the 72-pair frame (6 known positives), four standard classifiers, namely logistic regression (LR; max_iter=10,000), random forest (RF; 500 trees, seed 42), SVM with RBF kernel, and SVM with linear kernel, were trained on five features (|DD|, PCS, ΔExpression, necessity, mutation frequency), all with default scikit-learn hyperparameters and no tuning (production versions pinned in the repository Dockerfile; the audit suite is additionally verified under scikit-learn 1.9.0). The composite score was excluded from classifier inputs because it is a deterministic function of the other features. Features were z-score standardized within each training fold (standardization parameters estimated on the training pairs only and applied to the held-out pair, preventing information leakage); evaluation used leave-one-pair-out cross-validation, scoring each held-out pair with the out-of-fold decision function (or predicted probability), and AUROC was computed from the pooled out-of-fold scores with the rank-based Mann–Whitney statistic. With only 6 positive pairs these classifier results are noise-dominated and are reported as descriptive context, not as model selection.

1.5 Composite score definition and provenance

The composite score is a deterministic weighted combination of four min-max-normalized components: PCS (weight 0.50), |DD| (0.20), $-\log_{10}(q)$ (0.15), and mutation frequency (0.15), with normalization applied globally across the analysis table (pcs.py); ΔExpression is *not* a component. The weights are heuristic constants fixed in the analysis code; they were not fitted by nested resampling, and candidate-ranking iterations during development were evaluated against the same benchmark reported here. The composite AUROC (0.831) is therefore descriptive context for the value of combining interpretable components, not an independently validated model. Weight sensitivity: perturbing the weights with Dirichlet noise (concentration parameters = base weights $\times$ 20; 200 samples, seed 42) gave AUROC quartiles of 0.803–0.838 (median 0.819, full range 0.664–0.884), so the descriptive performance is not fragile to the exact weight choice (composite_weight_sensitivity.py).

1.6 CPTAC cohorts, missingness, and multiplicity

Seven CPTAC cohorts were analyzed (BRCA $n = 122$, COAD $n = 97$, LUAD $n = 110$, GBM $n = 99$, PDAC $n = 140$, UCEC $n = 95$, LUSC $n = 108$; 771 tumors in total). Pearson correlations of $\log_2$ protein abundance were computed per paralog pair within each cohort, requiring ≥ 10 paired samples per test; samples lacking either protein measurement were excluded pairwise (no imputation). Significance used Benjamini–Hochberg correction within each pair across the 7 cohorts. The mutation-conditioned UCEC analysis compared ARID1B protein abundance between ARID1A-mutant ($n = 37$) and wild-type ($n = 58$) samples with the Wilcoxon rank-sum test. The ovarian HGSC global proteome (PDC study PDC000360, $n = 232$) was downloaded for reference but not used in the main analysis.

1.7 PRISM analysis

The PRISM Repurposing secondary screen (1,482 compounds × 727 cell lines) was analyzed as driver-genotype–conditioned drug sensitivity: $\Delta\text{AUC} = \text{mean}(\log_2 \text{AUC in driver-mutant lines}) - \text{mean}(\log_2 \text{AUC in wild-type lines})$. Selectivity calls required BH $q < 0.25$, $\Delta\text{AUC} < 0$, and enrichment > 0.1; this deliberately permissive discovery-stage threshold admits up to ∼25% expected false positives among passing associations. Drug–genotype associations do not imply direct binding of the paralog protein. We did not adjust for cell-line proliferation rate, a known correlate of drug response in pooled viability screens; PRISM associations are therefore exploratory.

1.8 Dependency window score (DWS)

The DWS (Equation 2 of the main text) is the ratio of the signed numerator $\max(\text{DD}, 0)$, which credits only compensation-direction shifts, to a conservative baseline-essentiality denominator, $\max(|\mu_{\text{Chronos}}|,\ f_{\text{pan-essential}}$ where $f_{\text{pan-essential}}$ is the fraction of lines with paralog Chronos score < -0.5 and the 0.01 floor prevents division by near-zero values. Selectivity is the mutant-minus-wild-type difference in the fraction of lines dependent on the paralog. Pairs were classified as PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$), HIGH_SELECTIVITY (selectivity > 0.15 and DWS > 1.0), MODERATE, or LOW_SELECTIVITY across 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer) on the ≥5-mutant frame. Robustness was assessed by re-deriving rankings under alternative denominators ($|\mu_{\text{Chronos}}|$ only; $f_{\text{pan-essential}}$ only; their mean), alternative floors (0.001, 0.05), and alternative selectivity thresholds (0.10, 0.20), and by bootstrap 95% confidence intervals for mean DWS and selectivity, plus per-pair bootstrap rank intervals, from 1,000 stratified resamples of cell lines within each context (seed 42; Tables S5 and S10). The |DD|-numerator variant (pre-revision formula) is retained as a sensitivity analysis (Table S10).

1.9 Sequence- and structure-derived descriptors

Sequence features (length, molecular weight, pI, GRAVY hydrophobicity, disorder propensity, lysine/cysteine density) were computed from UniProt canonical sequences and domain architectures from UniProt annotations. For 13 of the 15 scored proteins, single-chain structures of the N-terminal 400 residues were predicted de novo with ESMFold v1 (mean pLDDT = 0.724); ARID1A and ARID1B (∼2,200 residues each) were assessed with sequence-level features only because the truncation covers only ∼17% of the chain. Three heuristics are emphasized as exploratory: the feature-space structural similarity $\max(0, 1 - \overline{|\Delta\text{feature}|})$ over the sequence features (not a 3D alignment metric); a sliding-window pocket count (windows with pLDDT > 0.7 and hydrophobic fraction 0.2–0.7, aggregated per 12 windows); and a PROTAC-suitability score (lysine density $\times 2.0$ + disorder propensity − cysteine density $\times 0.5$). The per-pair targetability score combined structural similarity (0.25), domain-architecture conservation (0.30), the druggability proxy (0.25), and the PROTAC score (0.20); the composite prioritization score combined max-normalized DWS (0.40), selectivity rescaled to $[0, 1]$ as $(s + 1)/2$ (0.30), and targetability (0.30), with all weights fixed heuristically before final reporting in `alphafold_analysis.py` and no calibration against drug-development data. Sequence identity was approximated by k-mer Jaccard similarity ($k = 3$), validated against Needleman–Wunsch global alignment identity on 50 protein pairs (Pearson $r = 0.88$; Fig. S9a); the ≥30% proxy threshold is an enrichment filter, not a precise identity cutoff.

1.10 External digenic screens (Harle 2025; Flister 2025)

Data and hit definitions. Harle et al. (Genome Biol 2025) screened 472 nominated synthetic-lethal candidate pairs with a dual-guide CRISPR/Cas9 library in 27 cell lines (10 melanoma, 9 lung, 8 pancreatic). We used their long-format genetic-interaction table (their Table S4; mean normalized GI score and FDR per pair–line) with their hit criterion (mean normalized GI < -0.5, FDR < 0.01, neither single-target depleted; a pair is a hit if hit in $\geq$1 line; 117/472 pairs). 26 of the 27 lines carry DepMap identifiers (line C092 excluded from genotype stratification). Flister et al. (Cell Rep 2025) screened 36,648 paralog pairs with digenic enAsCas12a libraries in NCI-H1299 and MDA-MB-231 and meta-analysed 462 pairs across 49 models. We used their per-model lethal calls (their Table S3; 117 and 51 lethal pairs per model, 22 shared, 146 in the union; table values were used where the preprint text differs) for layer 1 and their 49-model meta-analysis table (their Table S6; hit if dzL2FC ≤ -2, reproducing the study's own SL counts for 98.3% of pairs) for layer 2. Both datasets were downloaded on 2026-08-01 and analysed without modifying any frozen pipeline component; file hashes and origins are recorded in `output/external_holdout.json`.

Layer 1 (mutation-agnostic). Each pair was scored by max|DD| across both orientations and all lineages ($\geq$3 mutant / $\geq$3 wild-type lines) with the frozen in4mer scoring engine (fidelity: maximum absolute difference 2.8×10^{-16} across 413 pairs). max|DD| separated screen hits from non-hits in both datasets (Harle: AUROC = 0.619, 10,000-permutation $p = 0.0002$, 411 scored pairs of which 98 hits; median |DD| 0.202 in hits vs. 0.135 in non-hits. Flister union: AUROC = 0.696, $p = 0.0001$, 29,457 scored pairs of which 114 hits; median 0.165 vs. 0.105; per model 0.671 (NCI-H1299) and 0.744 (MDA-MB-231)). The Harle library is enriched for pairs nominated by earlier RNAi and digenic studies; the Flister pair selection is unbiased.

Layer 2 (genotype-stratified). Driver-mutation status was projected onto screen models using the pipeline's class-specific rules (our Table S6). Only three curated pairs per dataset retained any driver-mutant models, and no per-pair test remained significant after Benjamini–Hochberg correction within each dataset:

Dataset	Pair	Mutant (hits)	Wild-type (hits)	Fisher p	MW p
Harle	SMARCA4→SMARCA2	5 (0)	21 (3)	1.000	0.575
Harle	ARID1A→ARID1B	2 (0)	24 (1)	1.000	0.972
Harle	EP300→CREBBP	2 (0)	24 (9)	1.000	0.277
Flister	MAP2K1→MAP2K2	1 (0)	47 (8)	1.000	0.282
Flister	SMARCA4→SMARCA2	2 (0)	46 (5)	1.000	0.741
Flister	EP300→CREBBP	3 (2)	46 (20)	0.422	0.121

Pooled across evaluable pairs, digenic hit rates did not differ between driver-mutant and wild-type models (Harle: 0/9 vs. 27/121, Fisher $p = 1.0$, GI one-sided Mann–Whitney $p = 0.899$; Flister: 2/6 vs. 59/235, Fisher $p = 0.475$, Mann–Whitney $p = 0.150$). The single directionally consistent case is EP300→CREBBP in Flister (2/3 mutant models hit vs. 20/46 wild-type; mean GI -2.84 vs. -1.61). With at most 9 mutant model-observations per dataset, layer 2 is uninformative on the driver-conditioned proposition; both source studies report low genotype penetrance of digenic paralog SL. Per-pair–per-model values are provided in `output/external_holdout_harle_layer2_lines.csv` and `output/external_holdout_flister_layer2_models.csv`.

2 Supplementary Figures

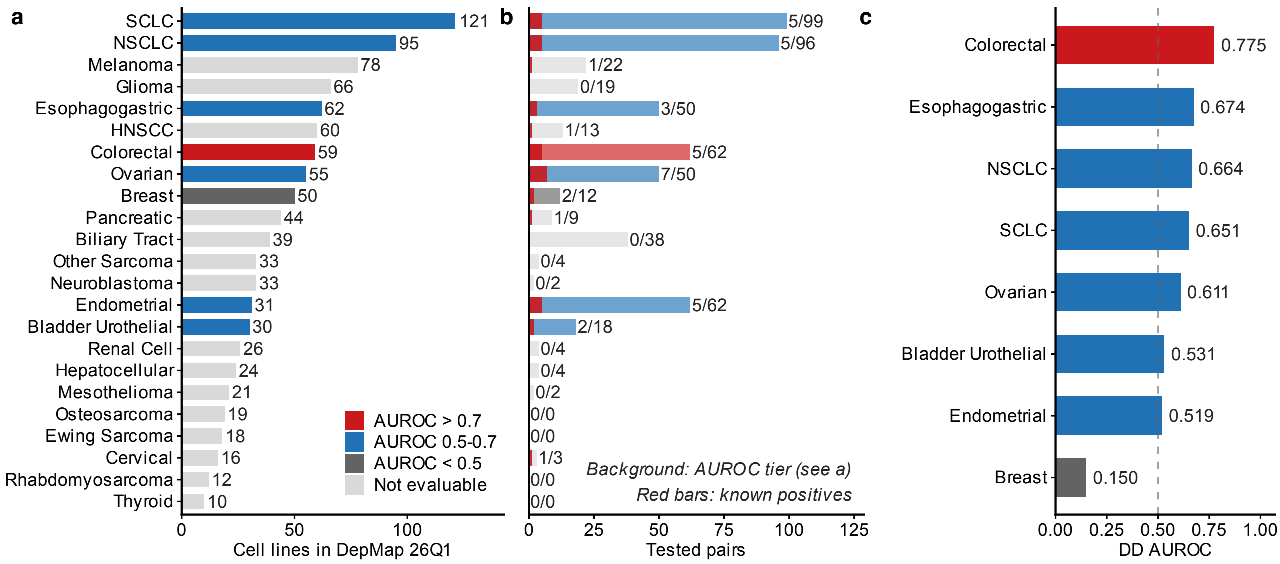

Fig. S1: Evaluation overview across 23 solid tumor types. a, Cell line counts per cancer type, colored by DD AUROC tier (red: AUROC > 0.7; blue: 0.5–0.7; gray: < 0.5; light gray: not evaluable). b, Tested paralog pairs per cancer type; red bars show known positives. Numbers indicate known/total pairs. c, DD AUROC for each of the 8 evaluable solid tumor types on the primary ≥5-per-group frame, colored by the same tier key as in a; bar labels show DD AUROC values. Values: Colorectal 0.775, Esophagogastric 0.674, NSCLC 0.664, SCLC 0.651, Ovarian 0.611, Bladder Urothelial 0.531, Endometrial 0.519, Breast 0.150. Dashed line, random classifier (0.5). Cancer types with <2 known positive pairs were excluded from AUROC evaluation (see Table S1 for the 15 non-evaluable types). On the ≥3-per-group sensitivity frame, four additional lineages become evaluable (Biliary Tract 0.990, Pancreatic 0.962, Melanoma 0.840, Cervical 0.714; 6 of 12 evaluable lineages exceed 0.7).

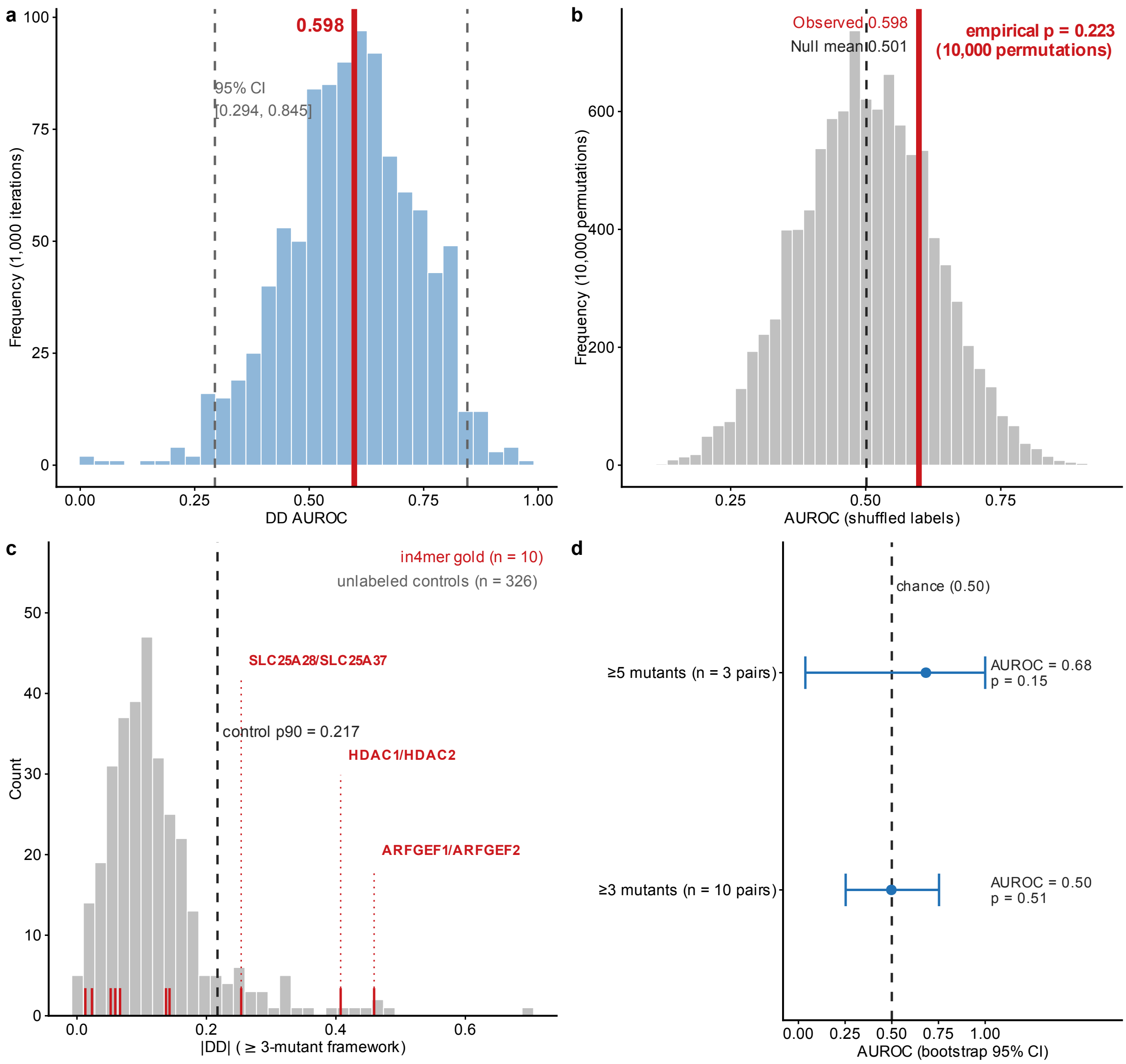

Fig. S2: Evaluation robustness: bootstrap, negative control, and external in4mer check. a, Bootstrap distribution of the per-pair DD AUROC on the max-aggregation frame (pair score = largest signed DD across the three evaluation lineages Ovarian/Endometrial/Cervical; 1,000 iterations; 72 pairs, 6 known positives; 95% CI: 0.294–0.845). b, Null distribution from 10,000 label-shuffled permutations on the same frame (empirical $p = 0.223$). The observed per-pair AUROC (0.598, max aggregation) does not significantly exceed the null mean (0.501): with only 6 positive pairs the aggregated per-pair evaluation remains underpowered, and the lineage-level evaluation (0.629, reported in the benchmark comparison) serves as the primary framework because it evaluates each driver×paralog×lineage combination separately rather than aggregating across lineages. c, External check against mutation-agnostic gold standards: distribution of |DD| (≥3-mutant framework) for 326 evaluable unlabeled control paralog pairs (gray histogram; 400 pairs sampled, seed 42). Red ticks: the 10 evaluable pairs of the 13 cross-platform in4mer gold standards (Esmaeili Anvar et al., Nat Commun 2024). Dashed line: control 90th percentile (0.217); the three pairs above it (ARFGEF1/ARFGEF2, HDAC1/HDAC2, SLC25A28/SLC25A37) are those with the most recurrent natural mutations. d, in4mer AUROC point estimates with bootstrap 95% CI for both evaluation frameworks. Only 3 of the 13 pairs meet the ≥5-mutant threshold (exploratory AUROC = 0.68); the 10 pairs evaluable at ≥3 do not separate from unlabeled controls (AUROC = 0.50, permutation $p = 0.51$), consistent with the mutation-agnostic, dual-knockout definition of these gold standards and with the limited power of a 10-positive comparison (main text, Limitations); adequately powered external tests against two 2025 digenic screens (Harle et al., Genome Biol 2025; Flister et al., Cell Rep 2025) appear in Supplementary Methods (External digenic screens).

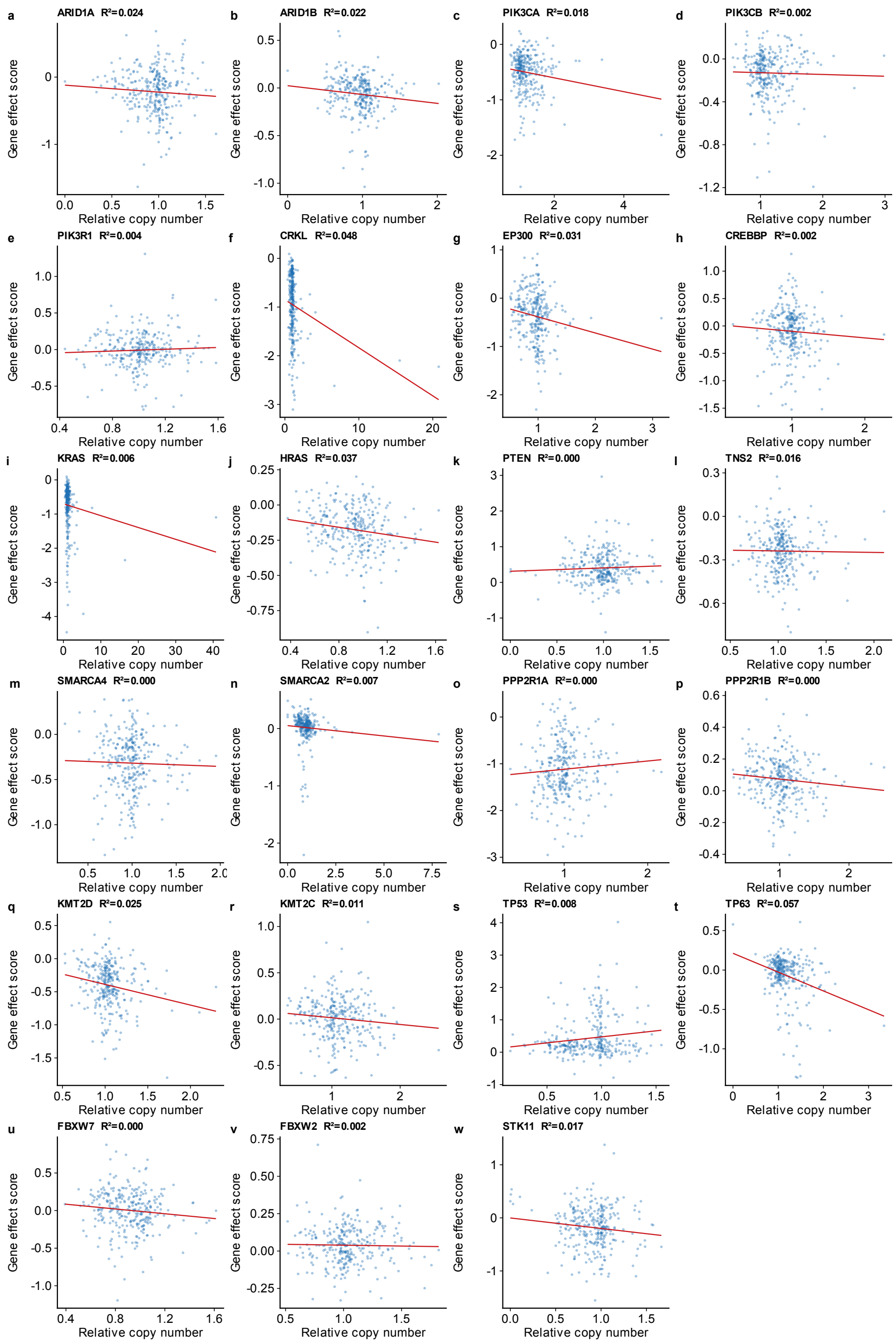

Fig. S3: CNV independence analysis. Scatter plots of relative copy number (x-axis) vs. Chronos gene-effect score (y-axis) for each of 23 paralog genes across 1,192 cell lines with both CNV and dependency data. Each panel shows one gene with the linear regression fit (red line) and R^2 value. All $R^2 < 0.10$, indicating that copy number variation explains <10% of gene-effect variance. Multivariate regression controlling for driver mutation and TP53 co-mutation left DD estimates essentially unchanged after CNV adjustment (see main text).

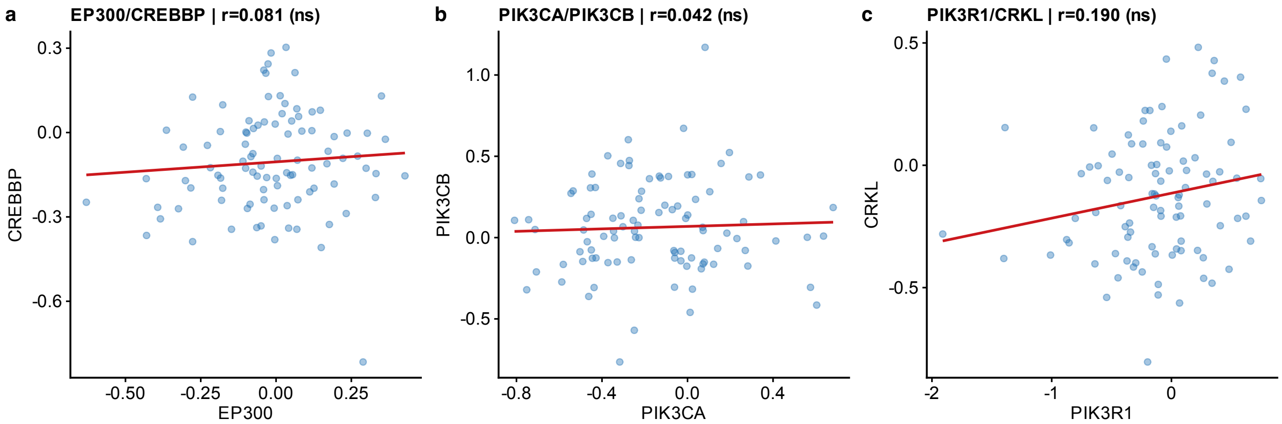

Fig. S4: CPTAC per-cohort scatter plots (UCEC). Driver vs. paralog protein abundance ($\log_2$ normalized intensity) in the UCEC cohort (95 tumors; $n = 83$ with both proteins quantified for a given pair): **a**, EP300 vs. CREBBP; **b**, PIK3CA vs. PIK3CB; **c**, PIK3R1 vs. CRKL. Each point represents one tumor sample. Line: linear regression fit with 95% confidence band. Pearson r is shown in each panel; significance stars: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$ (ns: BH-corrected $q \geq 0.05$). The complete set of per-cohort scatter plots for all 122 evaluable pair×cohort combinations is provided as a supplementary data archive in the public code repository (github.com/tjogzt/paralog-sl-predictor), with a machine-readable manifest.

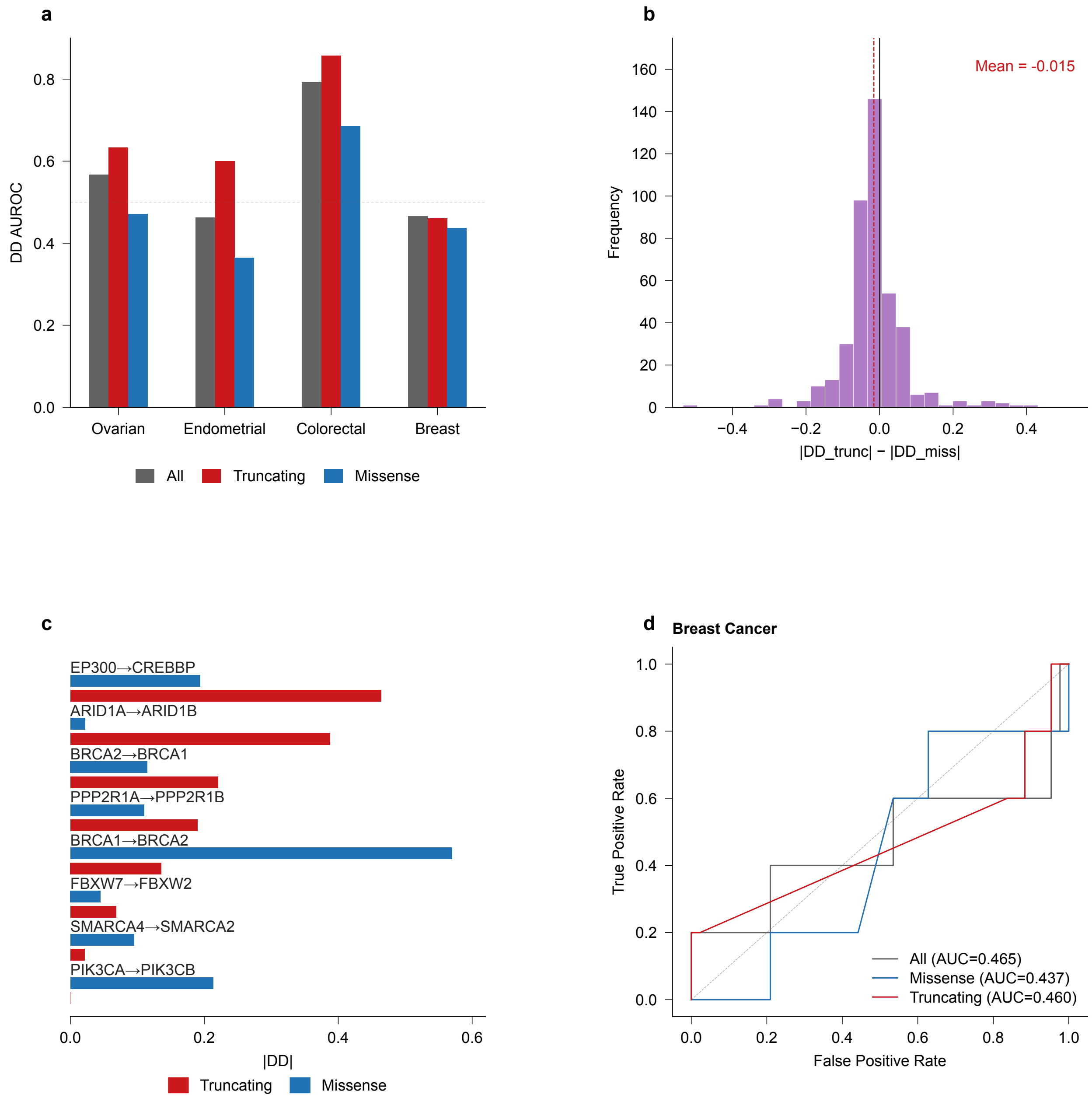

Fig. S5: Mutation-type stratification. a, DD AUROC stratified by mutation type (truncating vs. missense) across cancer types. b, Distribution of ΔDD (truncating minus missense) across all tested driver–paralog pairs; overall mean near zero (-0.015), indicating that truncating $>$ missense is pair-specific rather than universal. c, $|DD|$ under truncating (red) versus missense (blue) mutations for the curated SL pairs (maximum across cancer types, one row per pair); the truncating–missense gap is largest for ARID1A→ARID1B (+0.368 in Ovarian) and EP300→CREBBP (+0.314 in Colorectal). d, ROC curves for Breast cancer comparing truncating-only (AUROC = 0.460) vs. missense-only (0.437) and overall DD performance. Panel d is computed on the mutation-type analysis frame (48 Breast pairs); the resulting "All" AUC (0.465) therefore differs from the cross-cancer frame value (0.150; Fig. 1a), which uses a different pair universe.

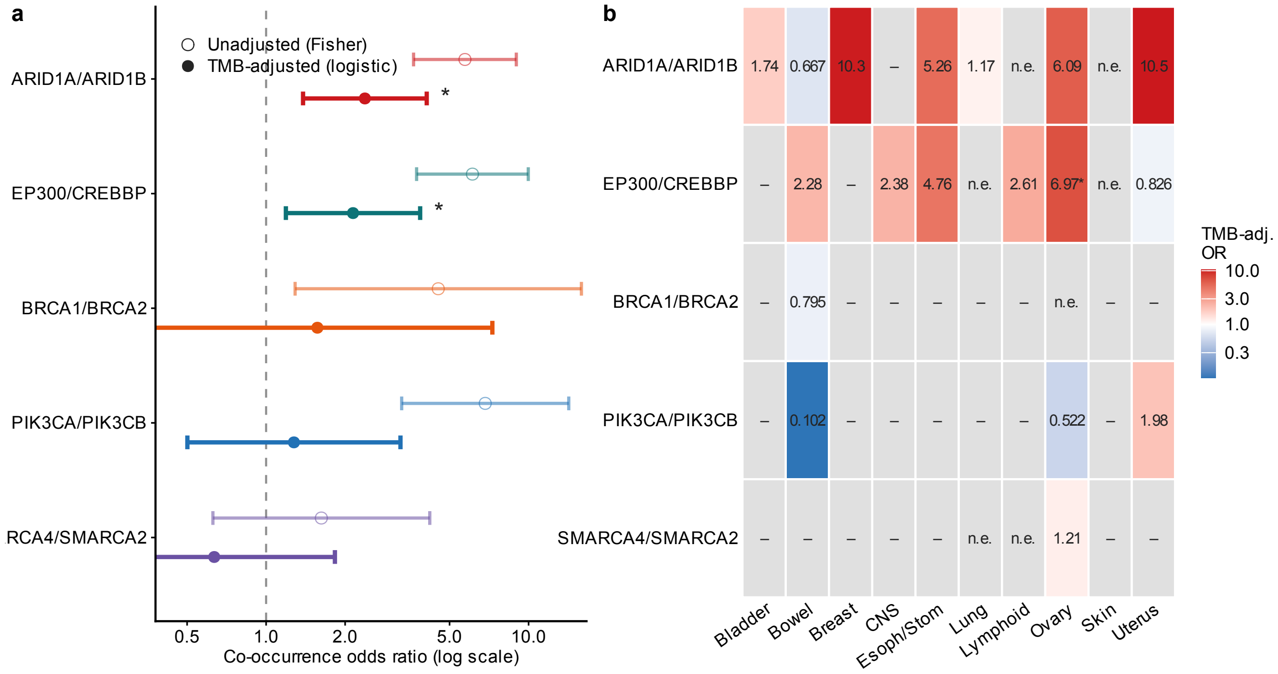

Fig. S6: Mutational co-occurrence of paralog pairs, with and without tumor mutational burden (TMB) adjustment. a, Pan-cancer co-occurrence odds ratios across 1,208 DepMap 26Q1 cell lines for the five testable pairs, unadjusted (Fisher's exact test, open circles) vs. TMB-adjusted (filled circles; logistic regression of paralog mutation status on driver mutation status and $\log_e(1+\text{TMB})$ per line). After TMB adjustment only ARID1A/ARID1B (adjusted OR = 2.38, $p = 0.002$) and EP300/CREBBP (adjusted OR = 2.14, $p = 0.011$) remain significant (*adjusted $p < 0.05$). b, Per-lineage TMB-adjusted odds ratios. "–": not evaluable (< 30 lines or < 3 mutant lines per gene within the lineage); "n.e.": not estimable (complete separation). Mutation status follows the gene-class-specific driver rules (Table S6). Source: `output/cooccurrence_by_lineage.json` (`cooccurrence_by_lineage.py`).

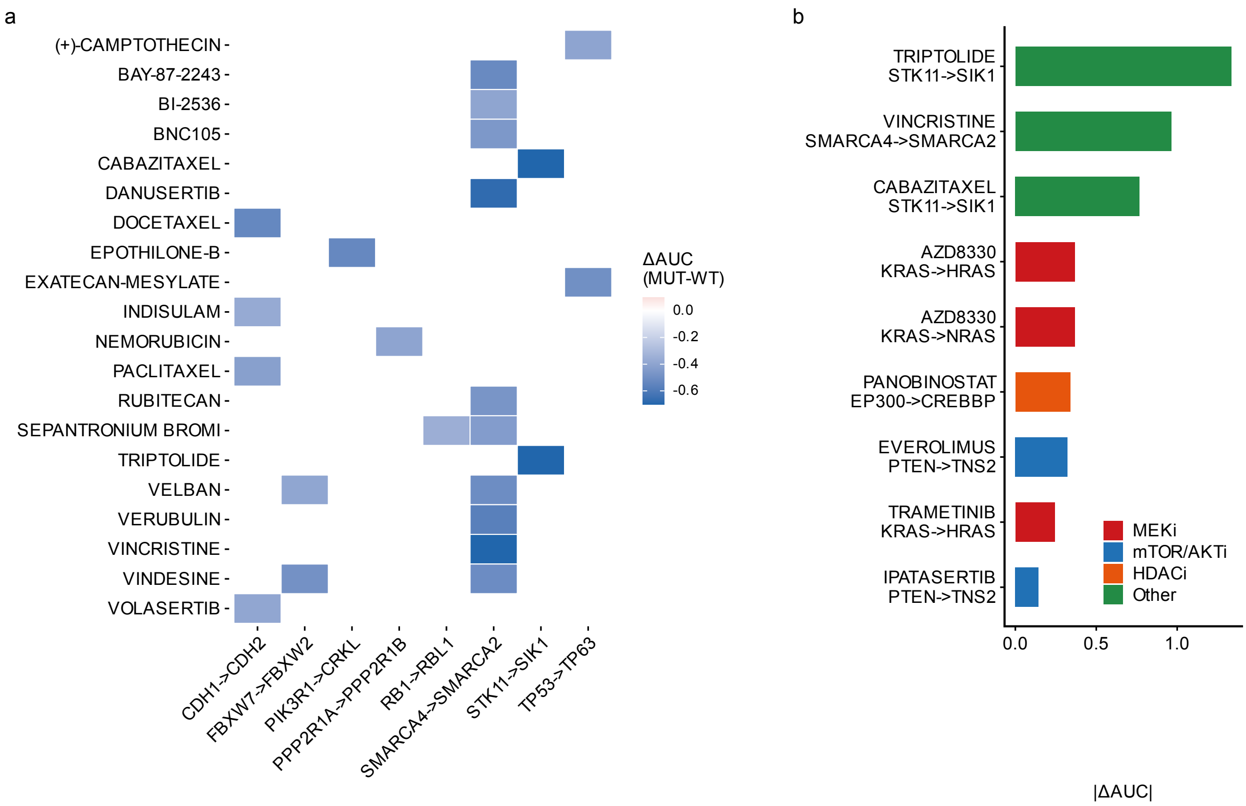

Fig. S7: PRISM drug selectivity and assay-validity anchors. **a**, Heatmap of drug sensitivity differential (ΔAUC = mean AUC in driver-mutant minus mean AUC in wild-type cell lines) for the top driver-genotype–drug sensitivity associations from the PRISM Repurposing screen (1,482 compounds, 727 cell lines). Negative ΔAUC indicates selective sensitivity in driver-mutant cells. Associations shown met discovery-stage criteria (BH $q <$ 0.25, $\Delta\text{AUC} < 0$, enrichment> 0.1). The displayed top associations are dominated by cytotoxic chemotherapies (tubulin inhibitors, topoisomerase poisons, and mitotic-kinase inhibitors) with $\Delta\text{AUC} < 0$, indicating genotype-conditioned cytotoxic sensitivity rather than paralog-specific targeting. **b**, Top driver-genotype–drug sensitivity associations ranked by $|\Delta\text{AUC}|$ and coloured by drug class (BH $q < 0.25$). Targeted agents recapitulate known driver-genotype biology and serve as assay-validity anchors: MEK inhibitors (AZD8330, Trametinib) selectively kill KRAS-mutant lines, mTOR/AKT inhibitors (Everolimus, Ipatasertib) kill PTEN-mutant lines, and the HDAC inhibitor Panobinostat kills EP300-mutant lines. The largest differentials are cytotoxic chemotherapies (“Other”), indicating genotype-conditioned cytotoxic sensitivity rather than paralog-specific targeting. Drugs shown are not necessarily direct binders of the paralog protein.

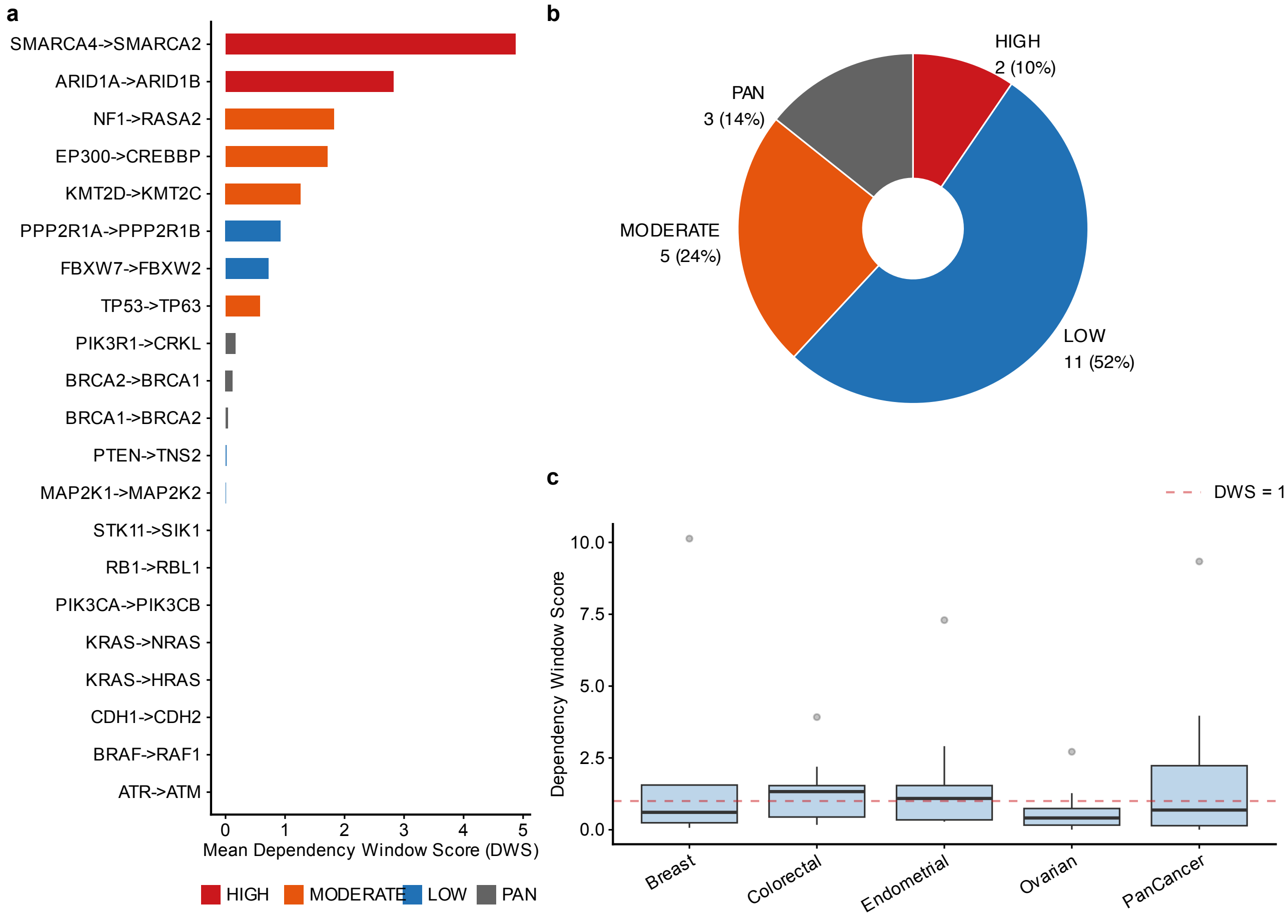

Fig. S8: Dependency window analysis. a, All 21 paralog pairs ranked by mean dependency window score (DWS) across up to 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer) on the ≥5-mutant frame. SMARCA4→SMARCA2 (4.87) and ARID1A→ARID1B (2.82) are the two HIGH_SELECTIVITY pairs; NF1→RASA2's signed DWS (1.82) derives entirely from the Endometrial context, where the 0.01 denominator floor is active, with selectivity ≈ 0 and a wide bootstrap rank interval (95% CI 1–18). b, In vitro selectivity tier classification: HIGH_SELECTIVITY (selectivity > 0.15 and DWS > 1.0; $n = 2$), MODERATE ($n = 5$), LOW_SELECTIVITY ($n = 11$), and PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$; $n = 3$). c, DWS values broken down by individual cancer context showing cross-lineage consistency for top candidates. The selectivity-versus-DWS classification view appears as main-text Fig. 4.

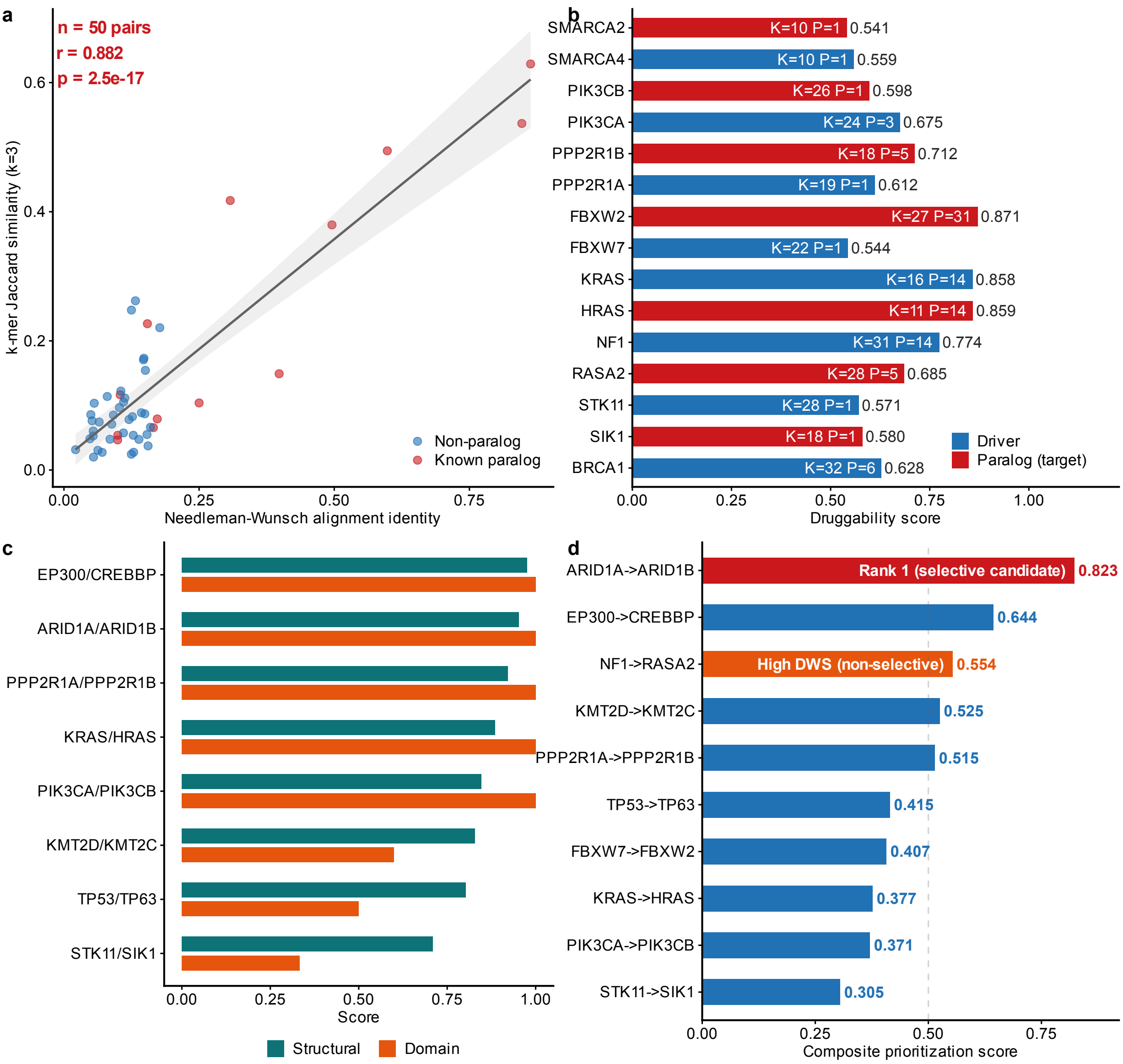

Fig. S9: Exploratory sequence- and structure-derived descriptors. a, Validation of the k-mer Jaccard proxy: scatter plot of k-mer Jaccard similarity ($k = 3$) vs. Needleman-Wunsch global alignment identity for 50 protein pairs (13 known paralogs, 37 non-paralog controls). Red: known paralog-SL pairs; blue: non-paralog pairs. Gray line: linear regression with 95% CI. b, Structure-based druggability assessment: composite druggability score for 15 proteins (7 paralog targets in red, 8 driver genes in blue) based on ESMFold-predicted structures. Scores integrate structural confidence (pLDDT), hydrophobic content, lysine density (K, relevant for PROTAC conjugation), and predicted pocket count (P). For 13 of the 15 proteins, ESMFold v1 structures of the N-terminal 400 residues were predicted de novo (mean pLDDT = 0.724); ARID1A and ARID1B (∼2,200 residues each) were assessed by sequence-level features only because the 400-residue truncation would cover only ∼17% of the chain (Supplementary Table S9). FBXW2 (0.871) and KRAS/HRAS (0.858–0.859) show the highest scores due to well-folded compact domains. c, Feature-space structural similarity (Structural) and UniProt domain-architecture conservation (Domain) for the eight pairs with the highest structural similarity among the ten pairs with non-zero domain overlap (of 13 scored pairs); five of the 13 (38%) show perfect domain conservation. The structural similarity is a heuristic, $\max(0, 1 - |\Delta\text{feature}|)$ over sequence-level features, not a three-dimensional alignment metric (Methods). d, Composite prioritization score (max-normalized signed DWS 0.40, rescaled selectivity 0.30, targetability 0.30; weights fixed heuristically before final reporting in `alphafold_analysis.py`). ARID1A→ARID1B ranks first (0.823), followed by EP300→CREBBP (0.644); NF1→RASA2 ranks third (0.554), its signed DWS resting on a floored denominator with selectivity ≈ 0. These descriptors are exploratory proxies and were not calibrated against drug-development data (main text, Methods and Limitations). Sources: `output/alphafold_structural_analysis.csv` and `output/druggability_analysis.json` (`R_figS9_descriptors.R`).

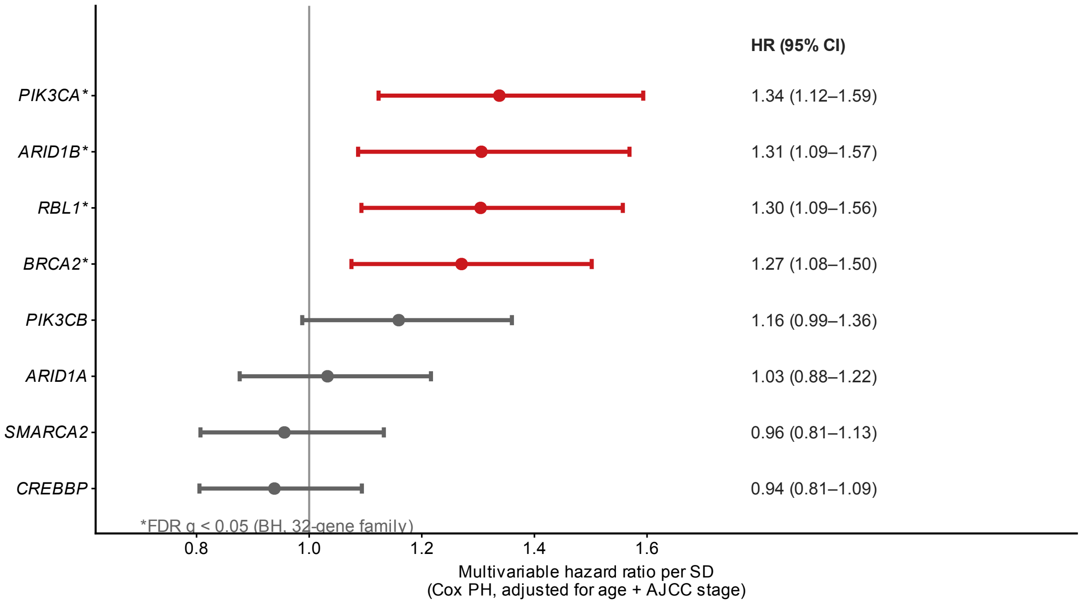

Fig. S10: Paralog expression and overall survival in TCGA BRCA. Forest plot of paralog gene expression vs. overall survival in TCGA PanCan Atlas BRCA (n = 1,069; Cox proportional-hazards on continuous $\log_2$-transformed, z-scored expression, adjusted for age and AJCC stage). Four genes withstand Benjamini–Hochberg correction across the 32-gene family (*PIK3CA*, $q = 0.035$; *ARID1B*, *BRCA2*, *RBL1*, each $q = 0.040$); these four are shown together with the compensating paralogs of the lead candidate pairs and *ARID1A* for direct contrast. Points, hazard ratio per SD; error bars, 95% CI; *marks FDR $q < 0.05$ (BH, 32-gene family). Full 32-gene results in Supplementary Table S8; UCEC and OV extensions in the Supplementary TCGA survival extension.

3 Supplementary Tables

3.1 Table S1: Cell line and evaluation summary across 23 solid tumor types

Label	Full name	Cell Lines	Pairs	Known+	DD AUROC
Colorectal	Colorectal adenocarcinoma	59	62	5	0.775
Esophagogastric	Esophagogastric adenocarcinoma / esophageal SCC	62	50	3	0.674
NSCLC	Non-small cell lung cancer	95	96	5	0.664
SCLC	Small cell lung cancer	121	99	5	0.651
Ovarian	Ovarian epithelial carcinoma	55	50	7	0.611
Bladder Urothelial	Bladder urothelial carcinoma	30	18	2	0.531
Endometrial	Endometrial carcinoma	31	62	5	0.519
Breast	Invasive breast carcinoma	50	12	2	0.150
Biliary Tract	Biliary tract cancer	39	38	0	—
Melanoma	Melanoma	78	22	1	—
Glioma	Diffuse glioma	66	19	0	—
HNSCC	Head and neck squamous cell carcinoma	60	13	1	—
Pancreatic	Pancreatic adenocarcinoma	44	9	1	—
Hepatocellular	Hepatocellular carcinoma	24	4	0	—
Renal Cell	Renal cell carcinoma	26	4	0	—
Other Sarcoma	Other soft-tissue sarcoma	33	4	0	—
Cervical	Cervical carcinoma	16	3	1	—
Mesothelioma	Pleural mesothelioma	21	2	0	—
Neuroblastoma	Neuroblastoma	33	2	0	—
Osteosarcoma	Osteosarcoma	19	0	0	—
Ewing Sarcoma	Ewing sarcoma	18	0	0	—
Rhabdomyosarcoma	Rhabdomyosarcoma	12	0	0	—
Thyroid	Thyroid carcinoma	10	0	0	—

Table S1: Cell line counts and DD AUROC across all 23 analyzed solid tumor types (primary $\geq$5-per-group frame). Top: AUROC > 0.7 (1 type). Upper middle: AUROC 0.5–0.7 (6 types). Lower middle: AUROC < 0.5 (1 type; Breast, where reverse-direction entries dominate under signed scoring). Bottom: insufficient known positives for AUROC (15 types). AUROC is computed on signed DD (primary metric). Known+: number of curated positive pairs evaluable in each lineage. "—": AUROC not computable (<2 known positives). Pairs: driver×paralog combinations passing the $\geq$5 mutant / $\geq$5 wild-type filter. On the relaxed $\geq$3-per-group sensitivity frame, four additional lineages become evaluable (Biliary Tract 0.990, Pancreatic 0.962, Melanoma 0.840, Cervical 0.714), giving 12 evaluable lineages in total (6 above 0.7). Full name: standardized cancer-type name corresponding to each short label; SCC, squamous cell carcinoma.

3.2 Table S2: Complete paralog-SL pair analysis results

The full analysis results are provided as a tab-separated file: `TableS2_FullResults.tsv` (110 rows $\times$ 17 columns, representing all driver×paralog×lineage combinations in the three evaluation lineages (Ovarian, Endometrial, Cervical) passing the minimum sample-size filter of $\geq$5 mutant and $\geq$5 wild-type cell lines). Effect sizes (Cohen's d, Hedges' g) and per-stratum Welch t-test p-values for the same 110 associations are provided as `TableS7_EffectSizes.tsv` (Supplementary Table S7).

Data dictionary:

- `driver_gene`: Driver gene with damaging mutation.
- `paralog_gene`: Candidate paralog tested for conditional dependency.
- `cancer_type`: Solid tumor lineage (DepMap OncotreePrimaryDisease).
- `pcs`: Paralog Compensation Score (DD $\times$ necessity).
- `delta_expression`: ΔExpression between MUT and WT groups.

- `necessity`: Mean gene-effect score of the paralog (more negative = more essential).
- `dependency_dd`: Delta Dependency (signed; positive = greater dependency in MUT).
- `composite_score`: Deterministic weighted composite of min-max-normalized PCS (0.50), |DD| (0.20), $-\log_{10}(q)$ (0.15), and mutation frequency (0.15); weights fixed heuristically in `pcs.py`. ΔExpression is not a component.
- `novelty`: "Known" if in the curated set (Table S3), else "Novel".
- `q_value`: Benjamini-Hochberg adjusted p-value within each driver gene.
- `mutation_frequency`: Fraction of cell lines with driver mutation in this lineage.
- `n_mut`, `n_wt`: Number of mutant and wild-type cell lines.
- `is_known_paralog_sl`: Boolean; TRUE if pair is in the curated set (Table S3).

3.3 Table S3: Curated paralog-SL pairs with evidence provenance and tier assignment

Tier	Driver	Paralog	Assay (perturbation)	Model system	Validated direction	Dual?	Indep.	Direct SL	Inclusion	Key ref
A	AKT1	AKT2	Combinatorial CRISPR digenic KO	Cancer cell lines	AKT1→AKT2	Yes	Yes	Yes	Primary	Najm 2018
A	CDK4	CDK6	Digenic KO (pgPEN library)	Cancer cell lines	CDK4→CDK6	Yes	Yes	Yes	Primary	Parrish 2021
A	MAP2K1	MAP2K2	Digenic KO (pgPEN library)	Cancer cell lines	MAP2K1→MAP2K2	Yes	Yes	Yes	Primary	Parrish 2021
B	SMARCA4	SMARCA2	CRISPR KO conditioned on natural SMARCA4 mutation	SMARCA4-mutant cancer lines	SMARCA4→SMARCA2	No	Yes	Conditional	Primary	Hoffman 2014
B	ARID1A	ARID1B	shRNA knockdown conditioned on natural ARID1A mutation	ARID1A-mutant cancer lines	ARID1A→ARID1B	No	Yes	Conditional	Primary	Helming 2014
C	EP300	CREBBP	p300 degradation / CRISPR in CREBBP-deficient lines	CREBBP-mutant cancer lines	Reciprocal only (CREBBP→EP300)	No	Yes	Reciprocal	Secondary	Ogiwara 2016; Nie 2021
C	PIK3CA	PIK3CB	shRNA / PI3K inhibitor in PTEN-deficient lines	PTEN-deficient cancer lines	PTEN→PIK3CB only	No	Yes	No (other driver)	Secondary	Wee 2008
C	CCNE1	CCNE2	Mouse developmental double knockout	Mouse embryo	CCNE1↔CCNE2 redundancy	Yes (mouse)	Yes	No (redundancy)	Secondary	Geng 2003
C	FBXW7	FBXW2	DepMap computational analysis	700+ cell lines	FBXW7→FBXW2	No	No	No (computational)	Secondary	DepMap 26Q1 release
C	PPP2R1A	PPP2R1B	DepMap computational analysis	700+ cell lines	PPP2R1A→PPP2R1B	No	No	No (computational)	Secondary	DepMap 26Q1 release
comp.	BRCA1	BRCA2	PARP inhibition / genetic screen (functional analogs)	BRCA-mutant cancer lines	BRCA1/2→PARP axis	No	Yes	Functional analog	Comparator	Bryant 2005
comp.	STK11	SIK1	Mouse genetics, LKB1–SIK axis (partial homolog)	Mouse NSCLC models	STK11→SIK1/3 axis	No	Yes	Pathway axis	Comparator	Hollstein 2019

Table S3: Twelve curated pairs with evidence provenance and tier assignment after verification of the primary citations. **Tier A** (3 pairs): direct genetic synthetic-lethal evidence from dual-gene perturbation (combinatorial/digenic CRISPR knockout). **Tier B** (2 pairs): natural-genotype conditional dependency, i.e., the paralog is a demonstrated selective dependency in driver-mutant cells, from single-gene perturbation with functional validation. The Tier A ∪ Tier B set (5 pairs) constitutes the primary literature-derived benchmark. **Tier C** (5 pairs): indirect evidence only (reciprocal-direction-only, other-driver, developmental-redundancy, or DepMap-derived); excluded from the primary benchmark and reported separately. **Comparators** (2 pairs): mechanistic reference pairs that are not sequence paralogs (functional analog or pathway axis); used as specificity references only. **Dual?**: whether the evidence comes from simultaneous dual-gene perturbation. **Indep.**: evidence independent of DepMap CRISPR data. **Direct SL**: whether the evidence directly demonstrates synthetic lethality in the direction scored in this study ("Conditional" = genotype-conditional dependency; "Reciprocal" = direct evidence for the reverse direction only). **Inclusion**: Primary = primary literature-derived benchmark; Secondary = reported separately; Comparator = specificity reference. All twelve pairs are HGNC gene-group sequence paralogs except the two comparators; validated directions refer to the driver→paralog direction scored here. Key ref entries are short names corresponding to the numbered reference list in the main text.

3.4 External combinatorial-CRISPR gold standards (in4mer)

Esmaeili Anvar et al. (2024) meta-analyzed five combinatorial CRISPR paralog screens (Dede 2020, Gonatopoulos-Pournatzis 2020, Parrish 2021, Thompson 2021, Ito 2021; full citations in the main text reference list) and defined 13 candidate cross-study gold-standard paralog synthetic lethals (paralog score $>$ 0.25, called a hit in more than one study): CNOT7–CNOT8, PITPNA–PITPNB, TIA1–TIAL1, SAR1A–SAR1B, PTP4A1–PTP4A2, GSK3A–GSK3B, CSNK2A1–CSNK2A2, CSNK1D–CSNK1E, MAPK1–MAPK3, ARFGEF1–ARFGEF2, HDAC1–HDAC2, ASF1A–ASF1B, and SLC25A28–SLC25A37. We assessed whether these pairs could serve as an external benchmark for DD. No in4mer pair involves a 40-panel driver gene; as the closest case we additionally audited MAPK1 (of MAPK1–MAPK3), which carries qualifying oncogenic hotspot mutations in zero DepMap 26Q1 cell lines (Table S6), so no in4mer gold-standard pair is evaluable in our driver-mutation-conditioned framework. This reflects a genuine division of scope rather than missing data: the in4mer gold standards measure background-independent paralog buffering under dual knockout, whereas DD quantifies dependency shifts conditioned on naturally occurring driver mutations. We therefore report the in4mer set as a conceptual cross-reference rather than a benchmark, and we note that only 8 of the 13 pairs replicated as hits in at least three of the four in4mer screens (Chou et al. 2025). Repeating the control-pair sampling of the $\geq$3-mutant in4mer frame across 20 seeds (400 control pairs per seed) gave AUROC point estimates of 0.480–0.522 (median 0.499), confirming that the null separation is not an artifact of the control-sampling seed (`in4mer_seed_sensitivity.py`). Adequately powered external tests against two recent digenic screens (Harle et al., Genome Biol 2025; Flister et al., Cell Rep 2025) are reported in Supplementary Methods (External digenic screens): mutation-agnostic |DD| separates screen hits from non-hits in both datasets.

3.5 Table S4: 40-gene driver panel

Gene	Cancer Types	Source	Category
TP53	Pan-cancer	TCGA+COSMIC	TSG
KRAS	Lung, Colorectal, Pancreatic, Endometrial	TCGA+COSMIC	Oncogene
PIK3CA	Breast, Endometrial, Cervical, Colorectal	TCGA+COSMIC	Oncogene
PTEN	Endometrial, Ovarian, Breast, Glioma	TCGA+COSMIC	TSG
ARID1A	Endometrial, Ovarian, Colorectal	TCGA+COSMIC	TSG
BRCA1	Ovarian, Breast	TCGA+COSMIC	TSG
BRCA2	Ovarian, Breast	TCGA+COSMIC	TSG
BRAF	Melanoma, Colorectal, Thyroid	TCGA+COSMIC	Oncogene
NF1	Glioma, Breast, Lung	TCGA+COSMIC	TSG
RB1	SCLC, Breast, Bladder	TCGA+COSMIC	TSG
EGFR	Lung, Glioma, HNSCC	TCGA+COSMIC	Oncogene
ERBB2	Breast, Esophagogastric	TCGA+COSMIC	Oncogene
CDH1	Breast, Gastric	TCGA+COSMIC	TSG
CTNNB1	Endometrial, Hepatocellular	TCGA+COSMIC	Oncogene
STK11	Lung	TCGA+COSMIC	TSG
KEAP1	Lung	TCGA+COSMIC	TSG
EP300	Cervical, Colorectal	TCGA+COSMIC	TSG
FBXW7	Endometrial, Cervical, Colorectal	TCGA+COSMIC	TSG
PPP2R1A	Endometrial	TCGA+COSMIC	TSG
PIK3R1	Endometrial	TCGA	TSG
KMT2D	Endometrial, HNSCC	TCGA+COSMIC	TSG
CCNE1	Ovarian	TCGA+COSMIC	Oncogene
SMARCA4	Lung, Ovarian	TCGA+COSMIC	TSG
APC	Colorectal	TCGA+COSMIC	TSG
SMAD4	Colorectal, Pancreatic	TCGA+COSMIC	TSG
ATM	Pan-cancer	COSMIC	TSG
ATR	Pan-cancer	COSMIC	TSG
VHL	Renal	TCGA+COSMIC	TSG
SETD2	Renal	TCGA+COSMIC	TSG
BAP1	Mesothelioma, Renal	TCGA+COSMIC	TSG
PBRM1	Renal	TCGA+COSMIC	TSG
NOTCH1	HNSCC, Esophageal	TCGA+COSMIC	TSG
IDH1	Glioma	TCGA+COSMIC	Oncogene
SPOP	Prostate, Endometrial	TCGA+COSMIC	TSG
NFE2L2	Lung, Esophageal	TCGA+COSMIC	Oncogene
MET	Lung, Renal	TCGA+COSMIC	Oncogene
ALK	Lung	COSMIC	Oncogene
GATA3	Breast	TCGA	TSG
MAP3K1	Breast	TCGA	TSG
MYC	Pan-cancer	COSMIC	Oncogene

Table S4: Complete 40-gene driver panel (machine-readable mirror: `TableS4_DriverGenes.tsv`). Selection criteria: membership in TCGA PanCanAtlas driver analyses (Bailey et al. 2018) or the COSMIC Cancer Gene Census (recurrently mutated cancer drivers); per-gene DepMap 26Q1 mutant-line counts under the final class-specific rules are given in Table S6 (CCNE1 contributes zero mutant lines).

3.6 Table S5: All 21 paralog pairs ranked by dependency window score

Driver	Paralog	Num.	Denom.	DWS	Select.	Class	DWS 95% CI	Select. 95% CI	Rank 95% CI
SMARCA4	SMARCA2	0.186	0.055	4.874	0.181	HS	3.09–7.59	0.07–0.32	1–2
ARID1A	ARID1B	0.270	0.126	2.819	0.277	HS	1.72–5.89	0.19–0.37	1–4
NF1	RASA2	0.018	0.027†	1.824	0.005	MOD	0.00–4.84	0.00–0.02	1–18
EP300	CREBBP	0.264	0.159	1.710	0.108	MOD	1.00–4.37	0.00–0.21	2–6
KMT2D	KMT2C	0.046	0.039	1.261	0.056	MOD	0.50–3.56	0.02–0.11	2–8
PPP2R1A	PPP2R1B	0.074	0.080	0.925	0.000	LS	0.00–2.22	0.00–0.00	3–17
FBXW7	FBXW2	0.025	0.037	0.720	0.000	LS	0.24–2.79	0.00–0.00	3–9
TP53	TP63	0.024	0.050	0.577	0.037	MOD	0.11–2.40	0.01–0.07	3–11
PIK3R1	CRKL	0.152	0.955	0.167	−0.161	PE	0.00–0.68	−0.36–0.05	7–17
BRCA2	BRCA1	0.053	0.557	0.118	−0.047	PE	0.04–0.33	−0.23–0.14	8–13
BRCA1	BRCA2	0.021	0.545	0.039	−0.080	PE	0.00–0.20	−0.30–0.14	9–18
PTEN	TNS2	0.004	0.248	0.014	−0.006	LS	0.00–0.14	−0.07–0.07	10–18
MAP2K1	MAP2K2	0.001	0.186	0.003	−0.034	LS	0.00–0.45	−0.04–−0.02	7–18.5
CDH1	CDH2	0.000	0.143	0.000	−0.086	LS	0.00–0.00	−0.10–−0.07	16.5–19
KRAS	HRAS	0.000	0.195	0.000	−0.040	LS	0.00–0.11	−0.08–−0.01	10–18.5
ATR	ATM	0.000	0.057	0.000	−0.027	LS	0.00–0.96	−0.07–0.00	6–18.5
PIK3CA	PIK3CB	0.000	0.112	0.000	−0.065	LS	0.00–0.07	−0.11–−0.01	12–18.5
RB1	RBL1	0.000	0.050	0.000	0.007	MOD	0.00–0.00	0.00–0.02	16.5–19
BRAF	RAF1	0.000	0.199	0.000	−0.178	LS	0.00–0.00	−0.24–−0.12	16.5–19
STK11	SIK1	0.000	0.044	0.000	−0.011	LS	0.00–0.51	−0.02–−0.01	8–18.5
KRAS	NRAS	0.000	0.193	0.000	−0.060	LS	0.00–0.17	−0.11–−0.01	10–18.5

Table S5: All 21 analyzed paralog pairs ranked by the signed dependency window score (DWS = mean $\max(\mathrm{DD}, 0)$ / denominator; Equation 2 of the main text) on the ≥5-mutant frame. Num.: DWS numerator, mean $\max(\mathrm{DD}, 0)$ across cancer contexts (only compensation-direction shifts contribute). Denom.: baseline-essentiality denominator $\max(|\mu_{\mathrm{Chronos}}|, f_{\text{pan-essential}}, 0.01)$; †marks the only pair whose signed DWS derives from a context in which the 0.01 floor is active (NF1→RASA2, Endometrial). Select.: mean selectivity (fraction mutant-essential minus fraction wild-type-essential). Class: HS=HIGH_SELECTIVITY, MOD=MODERATE, LS=LOW_SELECTIVITY, PE=PAN_ESSENTIAL. DWS 95% CI and Select. 95% CI: bootstrap 95% percentile intervals from 1,000 stratified resamples of cell lines within each cancer context (seed 42; `dws_robustness.py`); the DWS and Select. columns report the observed point estimates, not bootstrap means. Rank 95% CI: bootstrap 95% interval of each pair's DWS rank across the same resamples. Three pairs evaluable under the previous frame (AKT1→AKT2, CCNE1→CCNE2, CDK4→CDK6) do not meet the ≥5-mutant threshold in any context and are not listed. The |DD|-numerator variant (pre-revision formula) is reported as a sensitivity analysis (Table S10); classification thresholds are operational stratifiers, not validated clinical cutoffs.

3.7 Table S6: Gene-class-specific driver-mutation rules and variant classification

Gene	Class	Rule	Old	New	% ret.
ALK	ONC	Hotspot	118	12	10.2
APC	TSG	LikelyLoF	217	108	49.8
ARID1A	TSG	LikelyLoF	240	173	72.1
ATM	TSG	LikelyLoF	165	59	35.8
ATR	TSG	LikelyLoF	125	37	29.6
BRAF	ONC	Hotspot	212	162	76.4
BRCA1	TSG	LikelyLoF	101	34	33.7
BRCA2	TSG	LikelyLoF	159	68	42.8
CCNE1	ONC	Hotspot	22	0	0.0
CDH1	TSG	LikelyLoF	87	34	39.1
CDK4	ONC	Hotspot	27	8	29.6
CTNNB1	ONC	Hotspot	85	45	52.9
EGFR	ONC	Hotspot	110	19	17.3
EP300	TSG	LikelyLoF	206	107	51.9
ERBB2	ONC	Hotspot	70	20	28.6
FBXW7	TSG	LikelyLoF	116	81	69.8
GATA3	TSG	LikelyLoF	52	16	30.8
KEAP1	TSG	LikelyLoF	99	31	31.3
KMT2D	TSG	LikelyLoF	356	185	52.0
KRAS	ONC	Hotspot	254	233	91.7
MAP2K1	ONC	Hotspot	36	14	38.9
MAP3K1	TSG	LikelyLoF	74	14	18.9
MAPK1	ONC	Hotspot	13	0	0.0
MET	ONC	Hotspot	66	1	1.5
NF1	TSG	LikelyLoF	215	122	56.7
PIK3CA	ONC	Hotspot	182	153	84.1
PIK3R1	TSG	LikelyLoF	79	46	58.2
PPP2R1A	TSG	LikelyLoF	53	13	24.5
PTEN	TSG	LikelyLoF	199	176	88.4
RB1	TSG	LikelyLoF	185	150	81.1
SMAD4	TSG	LikelyLoF	119	80	67.2
STK11	TSG	LikelyLoF	97	66	68.0
TP53	TSG	LikelyLoF	1171	1140	97.4

Table S6: Driver-mutation definitions by gene class (DepMap 26Q1, default-entry profiles). TSG: mutant requires `LikelyLoF`; ONC: mutant requires `Hotspot`. Old: cell lines carrying any non-silent variant (pre-revision permissive definition); New: cell lines qualifying under the class-specific rule; % ret.: 100 × New/Old. The top VariantInfo terms per gene are at `output/driver_mutation_rules.csv`. CCNE1 and MAPK1 are amplification-driven oncogenes with no qualifying hotspot mutations. The table covers the 40-gene driver panel plus genes audited outside the panel (CDK4 and MAP2K1 for the curated benchmark; MAPK1 for the external in4mer analysis).

3.8 Table S8: TCGA BRCA overall-survival associations for the 32 paralog genes

		Univariable			Multivariable (age + AJCC stage)			
Gene	n	**HR (95% CI)**	p	q	**HR (95% CI)**	p	q	**PH** p
PIK3CA	1069	1.22 (1.03–1.46)	0.024	0.383	1.34 (1.12–1.59)	0.001	0.035	0.239
RBL1	1069	1.13 (0.96–1.34)	0.135	0.513	1.30 (1.09–1.56)	0.003	0.040	0.359
ARID1B	1069	1.30 (1.09–1.55)	0.004	0.129	1.31 (1.09–1.57)	0.004	0.040	0.271
BRCA2	1069	1.10 (0.93–1.29)	0.257	0.513	1.27 (1.08–1.50)	0.005	0.040	0.695
STK11	1069	0.90 (0.77–1.05)	0.167	0.513	0.85 (0.72–0.99)	0.038	0.210	0.151
RAF1	1069	1.10 (0.94–1.28)	0.253	0.513	1.19 (1.01–1.40)	0.039	0.210	0.924
PIK3CB	1069	1.12 (0.96–1.30)	0.144	0.513	1.16 (0.99–1.36)	0.071	0.323	0.793
RASA2	1069	1.03 (0.86–1.24)	0.741	0.891	1.19 (0.97–1.44)	0.089	0.328	0.047
BRCA1	1069	1.16 (0.99–1.36)	0.066	0.513	1.15 (0.97–1.36)	0.100	0.328	0.353
KRAS	1069	1.15 (0.97–1.35)	0.113	0.513	1.16 (0.97–1.38)	0.103	0.328	0.750
SMARCA4	1069	1.05 (0.90–1.24)	0.531	0.823	1.13 (0.97–1.33)	0.122	0.349	0.061
TNS2	1069	0.96 (0.82–1.11)	0.562	0.823	0.89 (0.77–1.04)	0.131	0.349	0.156
PPP2R1B	1069	1.01 (0.85–1.21)	0.876	0.916	1.12 (0.93–1.34)	0.237	0.583	0.570
ATR	1069	0.99 (0.84–1.17)	0.892	0.916	1.09 (0.92–1.29)	0.321	0.620	0.633
FBXW2	1069	1.10 (0.93–1.30)	0.254	0.513	1.09 (0.92–1.30)	0.331	0.620	0.011
FBXW7	1069	1.01 (0.86–1.19)	0.908	0.916	1.08 (0.92–1.27)	0.337	0.620	0.066
HRAS	1069	0.90 (0.76–1.06)	0.200	0.513	0.92 (0.78–1.09)	0.338	0.620	0.040
PPP2R1A	1069	1.09 (0.94–1.27)	0.254	0.513	1.08 (0.92–1.27)	0.360	0.620	0.483
ATM	1069	0.99 (0.84–1.17)	0.916	0.916	1.08 (0.91–1.28)	0.368	0.620	0.071
BRAF	1069	0.90 (0.76–1.06)	0.199	0.513	0.93 (0.78–1.10)	0.404	0.632	0.391
CREBBP	1069	0.96 (0.83–1.11)	0.566	0.823	0.94 (0.81–1.09)	0.415	0.632	5.5×10^{-4}
RB1	1069	1.10 (0.93–1.30)	0.276	0.520	1.06 (0.89–1.26)	0.500	0.727	0.365
KMT2D	1069	1.05 (0.88–1.25)	0.592	0.823	1.05 (0.88–1.25)	0.576	0.772	0.135
PTEN	1069	0.97 (0.83–1.14)	0.730	0.891	1.05 (0.88–1.25)	0.586	0.772	0.024
SMARCA2	1069	0.90 (0.76–1.06)	0.205	0.513	0.96 (0.81–1.13)	0.603	0.772	0.174
PIK3R1	1069	0.88 (0.75–1.05)	0.149	0.513	0.96 (0.81–1.14)	0.671	0.809	0.282
ARID1A	1069	1.03 (0.88–1.20)	0.752	0.891	1.03 (0.88–1.22)	0.701	0.809	0.002
NF1	1069	1.07 (0.90–1.28)	0.457	0.812	1.03 (0.86–1.24)	0.732	0.809	0.014
SIK1	1069	0.87 (0.74–1.02)	0.089	0.513	0.97 (0.83–1.14)	0.740	0.809	0.604
CRKL	1069	0.95 (0.79–1.13)	0.549	0.823	1.03 (0.86–1.23)	0.759	0.809	0.048
KMT2C	1069	1.01 (0.85–1.20)	0.873	0.916	1.02 (0.85–1.22)	0.833	0.860	0.022
EP300	1069	1.03 (0.86–1.23)	0.751	0.891	1.01 (0.85–1.21)	0.881	0.881	0.007

Table S8: Cox proportional-hazards associations between paralog gene expression (continuous, $\log_2$-transformed, z-scored; HR per +1 SD) and overall survival in the TCGA PanCan Atlas BRCA cohort (n = 1,069, 151 events). Multivariable models adjust for age and AJCC stage (ordinal I–IV; complete cases n = 1,050, 143 events). q: Benjamini–Hochberg FDR across the 32-gene family within each model class; four genes pass FDR$<$ 0.05 in the multivariable family (*PIK3CA*, *RBL1*, *ARID1B*, *BRCA2*) and none in the univariable family. PH p: Schoenfeld-type residual–time correlation test of the proportional-hazards assumption (approximation without variance scaling); genes with PH $p < 0.05$ should be interpreted with caution. Sorted by multivariable p. Source: `output/tcga_survival_v2.json` (`tcga_survival_v2.py`).

3.9 TCGA survival extension: UCEC and OV cohorts

The BRCA survival pipeline (continuous $\log_2$-transformed, z-scored expression; Cox proportional-hazards models with Wald confidence intervals; BH-FDR within each model family) was applied to the two benchmark-relevant gynecological cohorts, UCEC (`ucec_tcga_pan_can_atlas_2018`; n = 525, 87 events) and OV (`ov_tcga_pan_can_atlas_2018`; n = 299, 180 events). AJCC stage annotation is defined but entirely unpopulated for these two studies in cBioPortal (verified at both patient and sample level), so the age-only adjusted model is the adjusted estimate of record for these cohorts. *ARID1B*: UCEC univariable HR = 1.17 (95% CI 0.941–1.464, p = 0.155, BH q = 0.381), age-adjusted HR = 1.17 (p = 0.179); OV univariable HR = 1.26 (95% CI 1.092–1.461, p = 0.002, BH q = 0.054), age-adjusted HR = 1.21 (p = 0.013, q = 0.406). The BRCA association was directionally concordant in OV (nominal age-adjusted p = 0.013, q = 0.406, not

FDR-significant) but not in UCEC; in BRCA it reached FDR only after covariate adjustment (univariable $q = 0.129$; multivariable $q = 0.040$), so its significance is adjustment-dependent. ARID1A-mutation stratification in UCEC (any non-silent *ARID1A* mutation; mutant $n = 226$, 19 events; wild-type $n = 299$, 68 events): *ARID1B* HR = 1.24 (95% CI 0.756–2.045, $p = 0.391$) in the mutant stratum versus HR = 1.15 (95% CI 0.905–1.473, $p = 0.248$) in wild-type, directionally consistent in both strata but not significant, with limited power in the mutant stratum (19 events). Machine-readable results: `output/tcga_survival_ucec.json`, `output/tcga_survival_ov.json`, and `output/tcga_survival_ucec_ov_summary.json` (`tcga_survival_ucec_ov.py`). The BRCA cohort forest plot is shown in Supplementary Fig. S10.

3.10 Table S9: Top paralog-SL candidates ranked by composite prioritization score

Table S9: Top paralog-SL candidates ranked by the exploratory composite prioritization score (moved from the main text when the structural analysis was demoted to exploratory descriptor status). DWS: mean dependency window score across up to 5 cancer contexts on the ≥5-mutant frame. Class: HS = HIGH_SELECTIVITY, MOD = MODERATE, LS = LOW_SELECTIVITY, PE = PAN_ESSENTIAL. Struct: feature-space structural similarity (heuristic, not a 3D alignment metric; Methods). Target.: composite prioritization score (max-normalized signed DWS 0.40, rescaled selectivity 0.30, targetability 0.30; weights fixed heuristically before final reporting in `alphafold_analysis.py`). Known SL pairs marked with ⋆. *ARID1A→ARID1B ranks first under the signed-DWS composite; NF1→RASA2's earlier top score was driven by a near-zero pan-essential denominator (0.1% of cell lines) that inflated its |DD|-based DWS, and with selectivity ≈ 0 it is not a selective candidate. These descriptors were not calibrated against drug-development data; all candidates are computationally nominated and require experimental validation. The machine-readable TSV (`TableS9_CompositeScore.tsv`) additionally carries an `evidence_tier` column mapping each candidate to its curated evidence tier (Tier A/B/C or comparator per Table S3); candidates outside the curated set are labelled `unlabeled`.*

Driver	Paralog	DWS	Class	Known	Struct	Target.
ARID1A	ARID1B	2.82	HS	⋆	0.952	0.823
EP300	CREBBP	1.71	MOD	⋆	0.976	0.644
NF1	RASA2	1.82	MOD		0.658	0.554
KMT2D	KMT2C	1.26	MOD		0.829	0.525
PPP2R1A	PPP2R1B	0.93	LS	⋆	0.921	0.515
TP53	TP63	0.58	MOD		0.803	0.415
FBXW7	FBXW2	0.72	LS	⋆	0.666	0.407
KRAS	HRAS	0.00	LS		0.885	0.377
PIK3CA	PIK3CB	0.00	LS	⋆	0.847	0.371
STK11	SIK1	0.00	LS	⋆	0.710	0.305

3.11 Table S10: Sensitivity of dependency-window score rankings and selectivity classification

Denominator variant	**Spearman ρ vs. base**	**Top-5 overlap**
Base: signed $\max(\mathrm{DD}, 0)$ numerator (production)	1.000	5
$\lvert\mu\rvert$ only	0.992	5
f (pan-essential fraction) only	0.966	3
$\mathrm{mean}(\lvert\mu\rvert, f)$	0.994	4
Floor 0.001	0.996	5
Floor 0.05	0.979	4
$\lvert\mathrm{DD}\rvert$ numerator (pre-revision formula, sensitivity)	0.381	4
Winsorized denominator (1%/99%)	0.999	5
Floor-dominated pairs removed	1.000	4
HIGH_SELECTIVITY selectivity threshold	**HS pairs**	**Flips vs. 0.15**
Selectivity > 0.10	3	1
Selectivity > 0.15 (base)	2	0
Selectivity > 0.20	1	1

Table S10: Sensitivity of DWS-based rankings to the numerator and denominator definitions (upper block) and of the HIGH_SELECTIVITY classification to the selectivity threshold (lower block), for the 21 evaluated paralog pairs. The production denominator is $\max(\lvert\mu\rvert, f, 0.01)$; upper-block rows vary the denominator, floor, or numerator as indicated. The winsorized-denominator row clips the denominator at its 1st/99th percentiles across all pair×context rows; the floor-dominated row removes the single pair whose denominator touches the 0.01 floor in at least one context (NF1→RASA2) and re-ranks the remaining 20 pairs, whose relative order is unchanged by construction (NF1→RASA2 drops out of the top five; PPP2R1A→PPP2R1B enters). Spearman ρ: rank correlation of the DWS ranking against the production definition; top-5 overlap: number of shared pairs in the top five of both rankings. At threshold 0.10, EP300→CREBBP additionally enters HIGH_SELECTIVITY; at 0.20, SMARCA4→SMARCA2 drops out; ARID1A→ARID1B remains HIGH_SELECTIVITY at all three thresholds. Source: `output/dws_robustness.json` (`dws_robustness.py`; 1,000 stratified bootstrap resamples, seed 42).

3.12 Table S11: Full regression model table for confounder controls

The complete regression results underlying the confounder-control analyses (main text, "Potential confounders and causal interpretation") are provided as a tab-separated file: `TableS11_RegressionModels.tsv` (43 rows × 14 columns; 11 driver–paralog pairs × up to four model variants). Models: `base` (gene effect ∼ driver mutation), `cnv_adj` (+ paralog copy number), `expr_adj` (+ paralog expression), `lineage_adj` (+ lineage fixed effects). Columns: `pair`, `model`, `n_mut/n_wt` (mutant and wild-type cell lines), `beta_mut` (mutation coefficient; DD = −`beta_mut`), `dd` (DD on the manuscript sign convention), `se` (HC3 heteroskedasticity-robust standard error), `ci_lo/ci_hi` (95% confidence interval), `p` (nominal p-value), `n` (cell lines in model), `q_bh` (Benjamini–Hochberg q within each model family), `se_lineage_cluster/p_lineage_cluster` (standard error and nominal p-value with cell lines clustered by Oncotree primary disease; addresses within-lineage correlation that HC3 cannot). One pair lacks a `cnv_adj` model because its copy-number covariate had insufficient variance. TP53 co-mutation control for ARID1A→ARID1B is reported separately in `output/regression_controls.json`. Source: `output/regression_table_full.csv` (`regression_controls.py`).

3.13 Table S12: Contextual literature values under non-comparable evaluation settings

Table S12: Contextual literature values under non-comparable evaluation settings. Published values are CV3 (gene-pair isolation) AUROCs from Feng et al. (2024; main-text reference list), Supplementary Data 1 (NSMRand negative sampling, 1:1 positive:negative ratio, complete dataset); individual method citations appear in the main-text reference list. The DD value from this study was evaluated on the full lineage-level frame (110 driver–paralog–lineage entries, 8 positives). *Evaluation frameworks differ: published methods were tested on general SL gene-pair universes with different datasets, tasks, and positive:negative ratios; DD was tested on paralog-SL pairs only. Direct AUROC comparison is not warranted.*

Method	CV3 AUROC
DD (this study)	0.629
SLMGAE	0.790
NSF4SL	0.683
GCATSL	0.678
GRSMF	0.656
PiLSL	0.626
KG4SL	0.563
SLGNN	0.530
PTGNN	0.529